



Do They Meet?

Quick facilitation notes & answer key

Goal

Students run two concurrent scripts and decide a combined result: do the two sprites land on the SAME number (meet) or not? Judging 'meet / don't meet' depends on holding BOTH parallel landings in mind at once — the Grade-2 C3.2 reasoning about how concurrent code behaves together. Here the scripts were written so the sprites DO meet (both land on 2).

Ontario curriculum (Grade 2 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential and concurrent events.

C3.2 — read and alter existing code, including code that involves sequential and concurrent events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: two counters (green for Bit, purple for Pixel) on a floor number path 0-5; after 'green flag!', see whether both counters end on the same square.

Run it (about 20 min)

1. I DO: read Bit's script — 'start 1, forward 2 → 3, back 1 → 2.' Bit lands on 2.
2. WE DO: read Pixel's script — 'start 4, back 1 → 3, back 1 → 2.' Press the green flag so BOTH run together.
3. YOU DO: Exercise 3 — students see both land on 2, so they circle YES, they meet.
4. Connect: 'They started in different places but both scripts ended on the same number — so they meet.'

Answer key (modelled by the run-gate)

Ex 1 — Bit: 1 → forward 2 → 3 → back 1 → 2. Bit lands on 2.

Ex 2 — Pixel: 4 → back 1 → 3 → back 1 → 2. Pixel lands on 2.

Ex 3 — both start on the green flag and both land on 2 → SAME number → YES, they MEET.

Interaction note: the scripts run independently and at the same time; we compare the two final positions to decide if they meet.

Watch for & differentiate

Common error: assuming sprites that start in different places can never meet — they can, if their scripts bring them to the same number.

Common error: running one script, then the other, as if sequential — the green flag starts BOTH at once (concurrent).

Simplify: find each landing first, then just ask 'are these two numbers the same?'

Extend: ask students to change ONE move in Pixel's script so the sprites would NOT meet.

Success indicator: the child lands both on 2 and circles YES, explaining the two scripts ended on the same number.